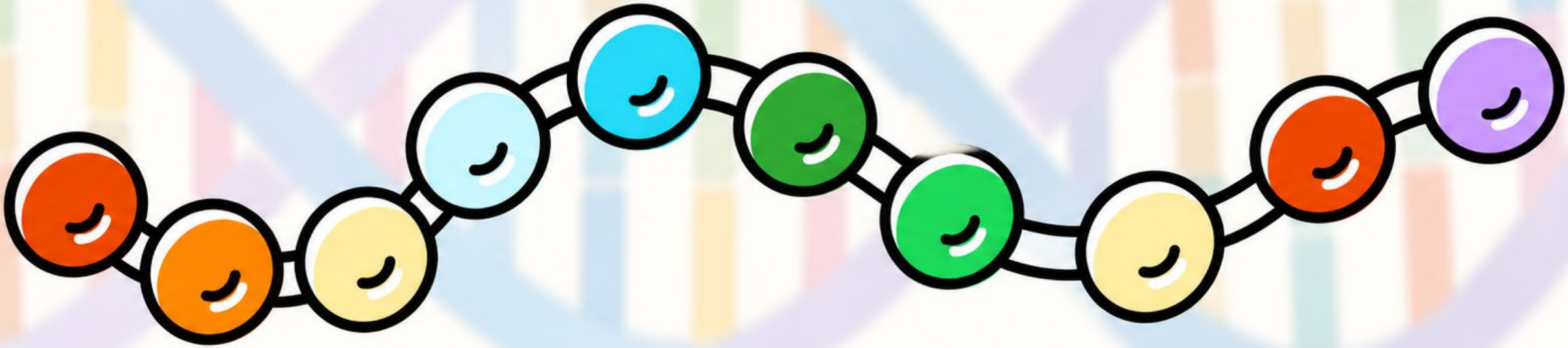
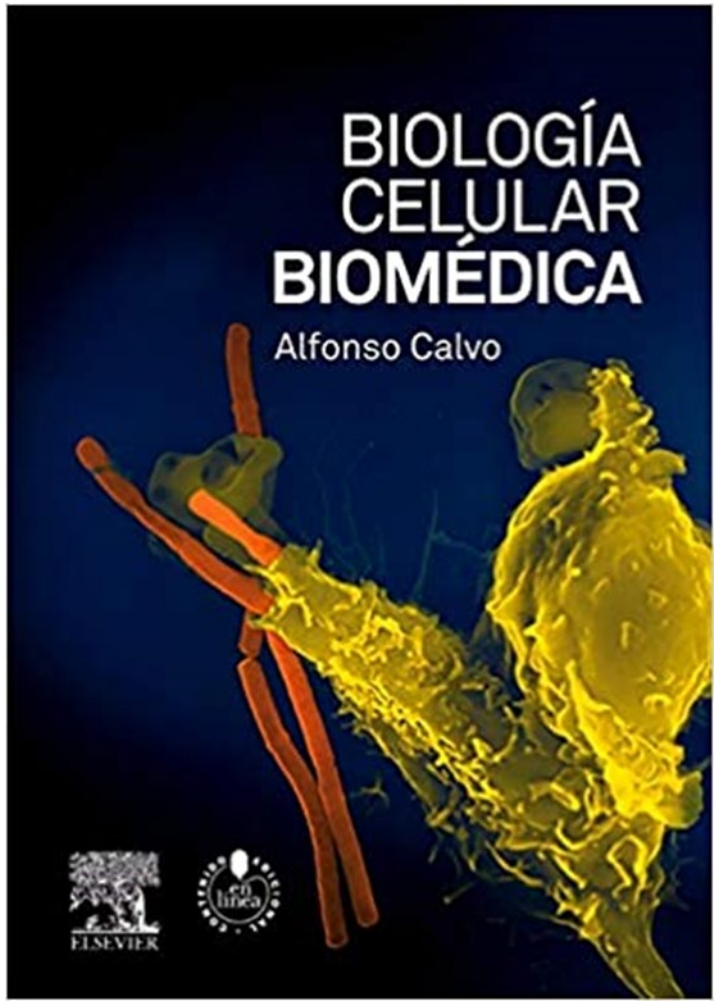


Protein synthesis

The genetic code in action

Supporting material



Reading/Study material

Calvo's BCB Chapter 9
(translated text in ADI)

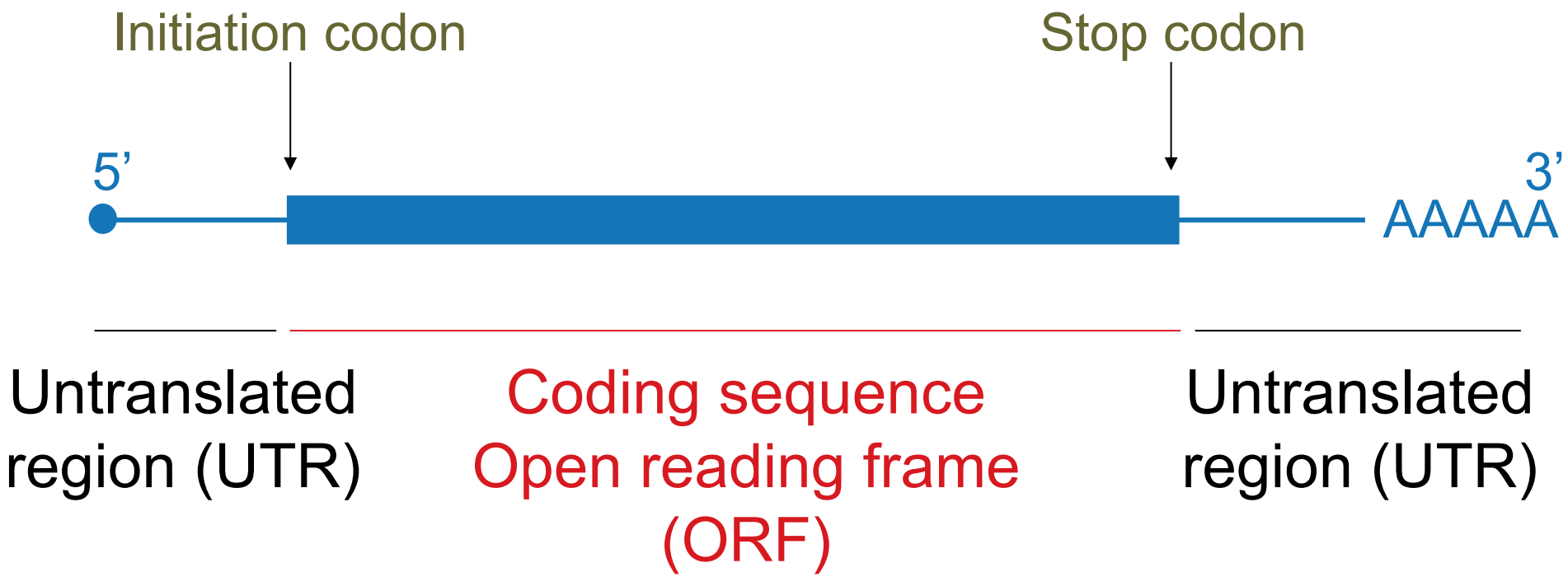
Learning goals

- To understand how the genetic information stored in DNA is transferred to an amino acid sequence to form a functional protein
- To know the basic molecular aspects of protein synthesis and the elements involved in the process
- To learn about some compounds that inhibit protein synthesis

Protein synthesis

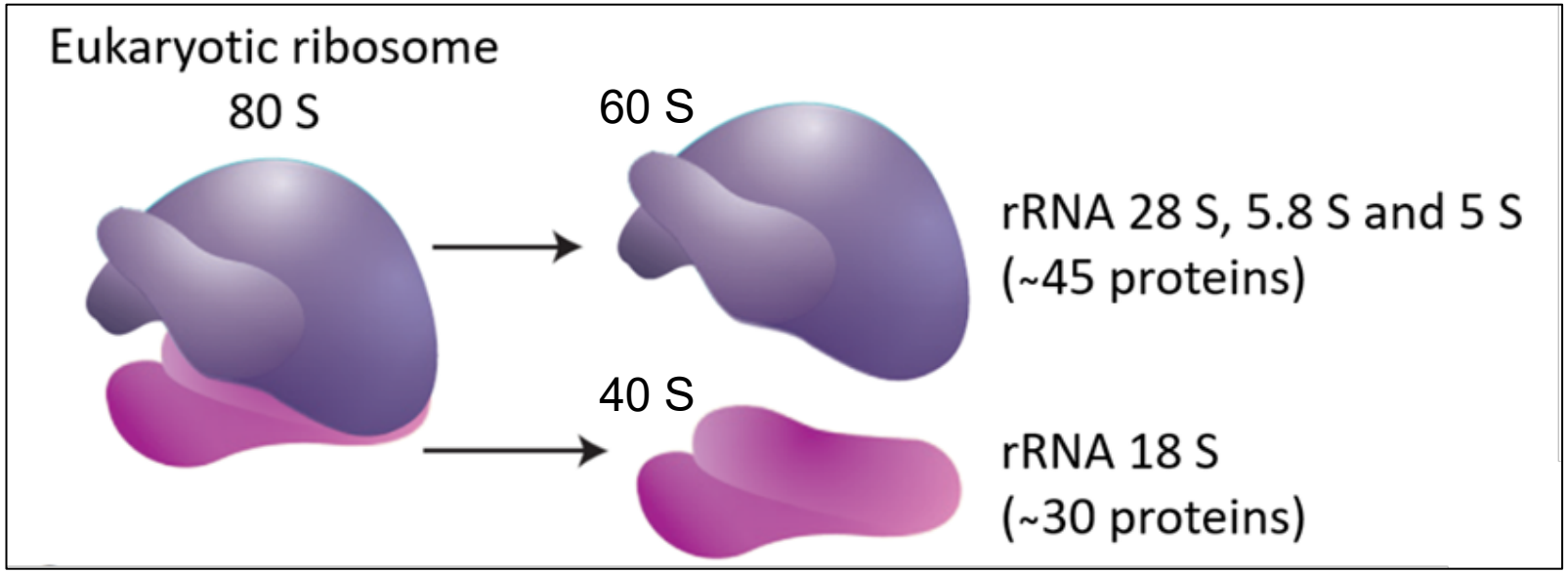
- Protein synthesis can use 90% of the chemical energy of a cell
- Possibly the most complex biosynthetic process (with more than 300 biomolecules involved)
    - Messenger RNA (mRNA)
    - Transfer RNA (tRNA)
    - Ribosomal RNA (rRNA)
    - Ribosomal proteins
    - Enzymes
    - Protein factors

mRNA

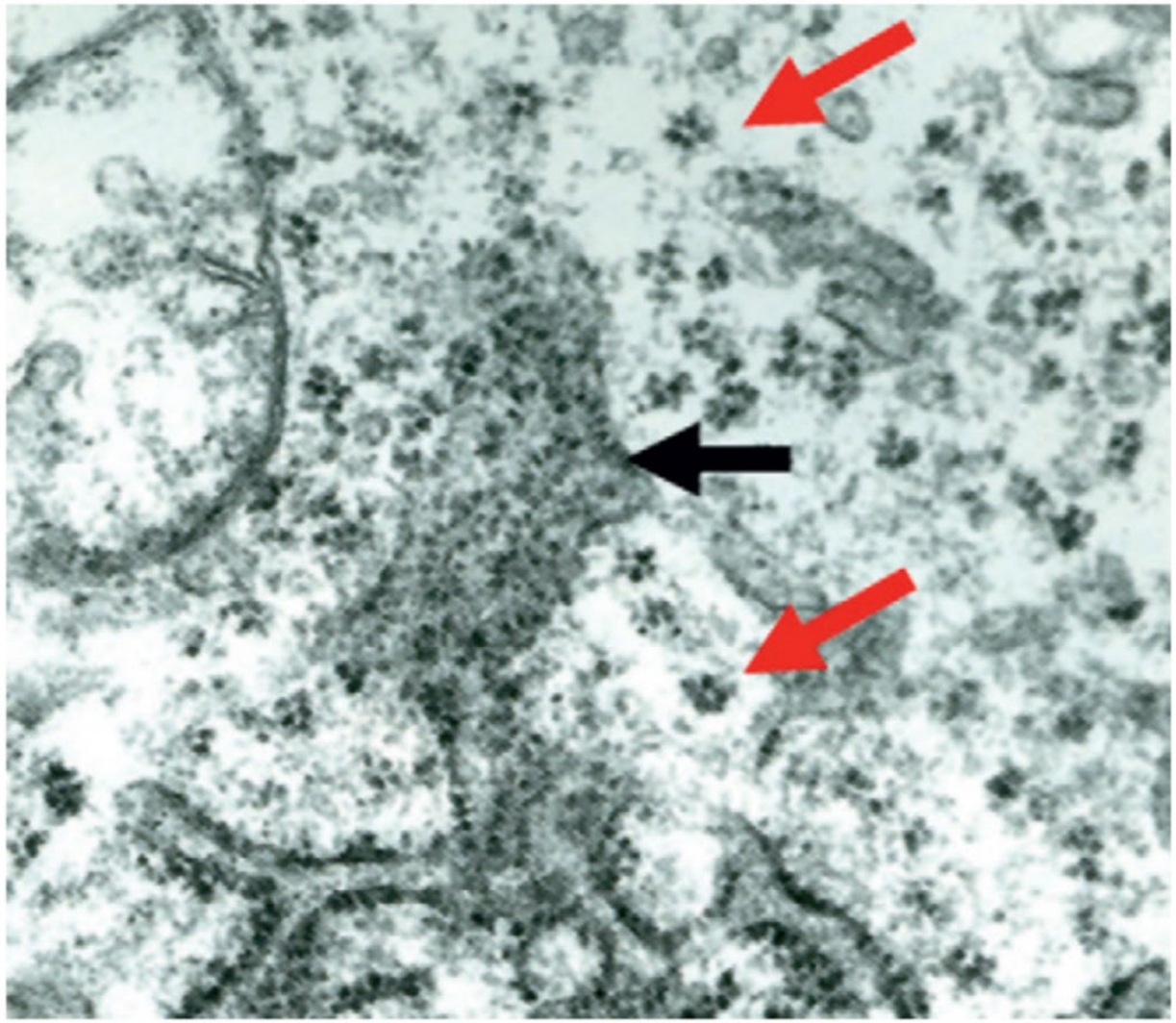

Initiation codon
Stop codon
5’
3’
AAAAA
Untranslated region (UTR)
Coding sequence Open reading frame (ORF)
Untranslated region (UTR)

Ribosomes

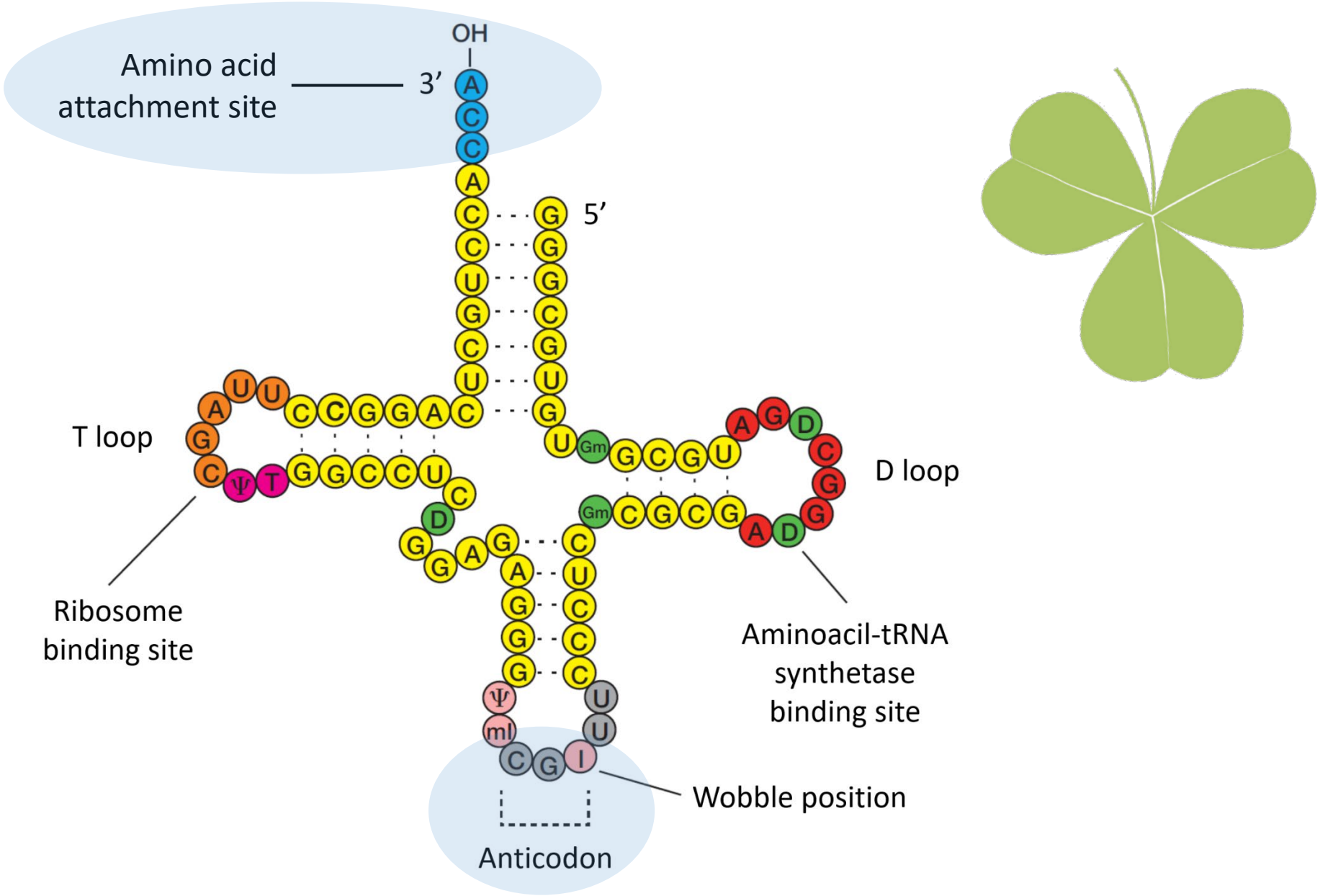
- Site of protein synthesis
- Supramolecular structures that that can account for around 25% of the dry weight of cells with high protein synthesis activity
- They are composed of rRNA (~65%) and ribosomal proteins (~35%)
- They are made of two unequal subunits bound together



Ribosomes are free in the cytosol or attached to the ER



tRNA


Amino acid attachment site
OH
3’
5’
T loop
D loop
Ribosome binding site
Aminoacil-tRNA synthetase binding site
Wobble position
Anticodon

Stages of protein synthesis

1) Activation of amino acids

   Aminoacyl-tRNA synthesis

Translation

2) Initiation

   mRNA and aminoacylated tRNA bind to ribosome

3) Elongation

   Cycles of aminoacyl-tRNA binding and peptide bond formation

4) Termination

   mRNA and protein dissociate, ribosomes are recycled

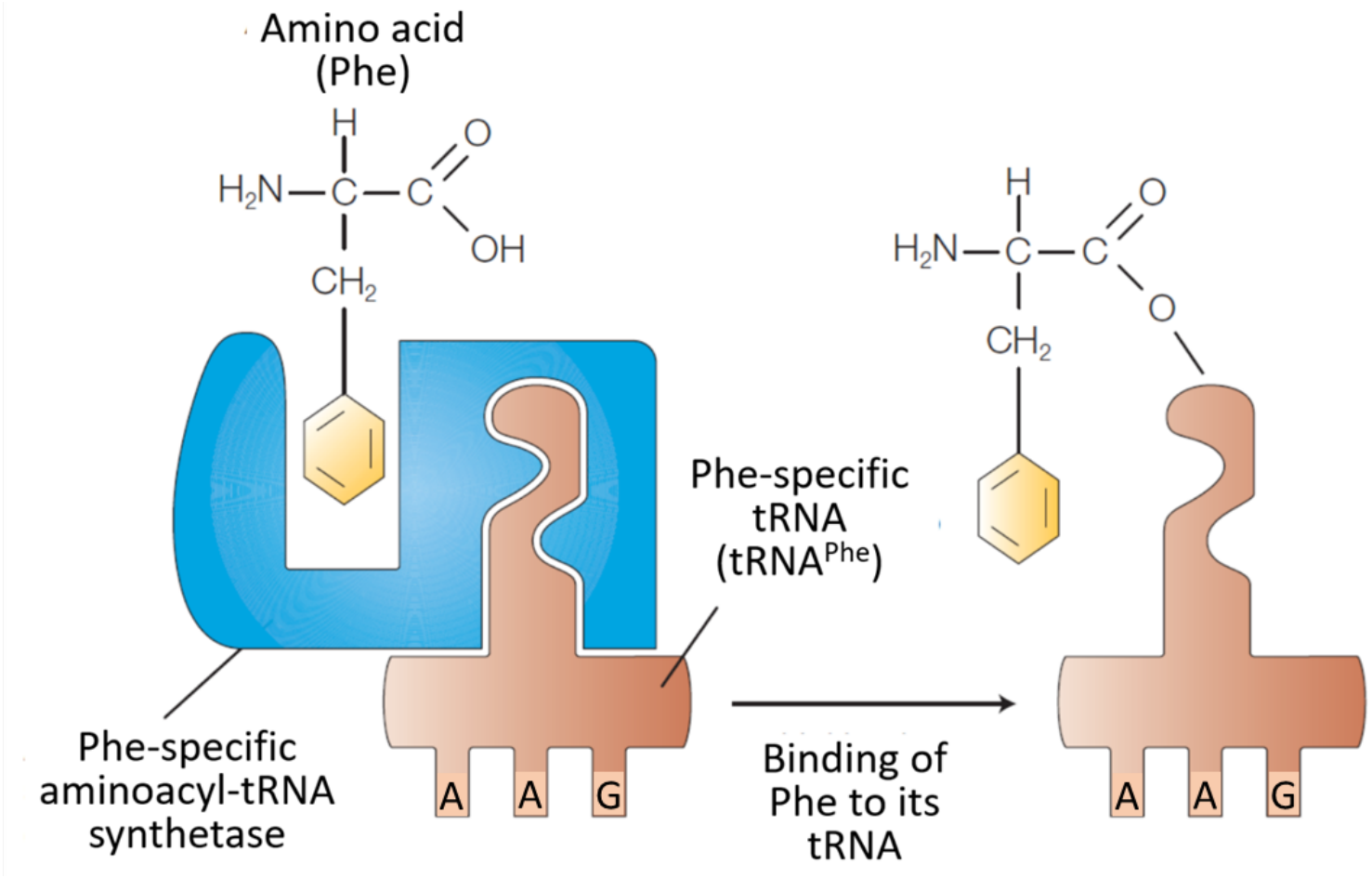
Aminoacyl-tRNA synthesis

- Aminoacyl-tRNA synthetases catalyze the esterification of amino acids to their corresponding tRNAs

$$\text{Amino acid} + \text{tRNA} + \text{ATP} \rightarrow \text{aminoacyl-tRNA} + \text{AMP} + PP_i$$

- Aims:
  - Activation of amino acids
  - Assignment of the correct amino acid to each codon in the mRNA

Aminoacyl-tRNA synthesis



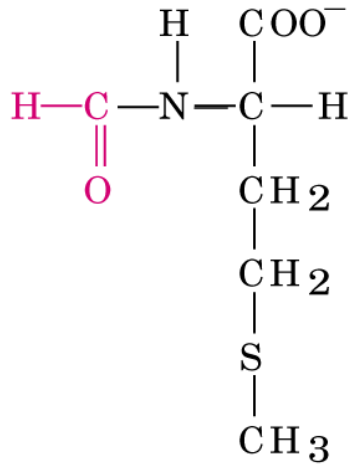
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https://www.youtube.com/watch?v=5bLEDd-PSTQ

Initiation

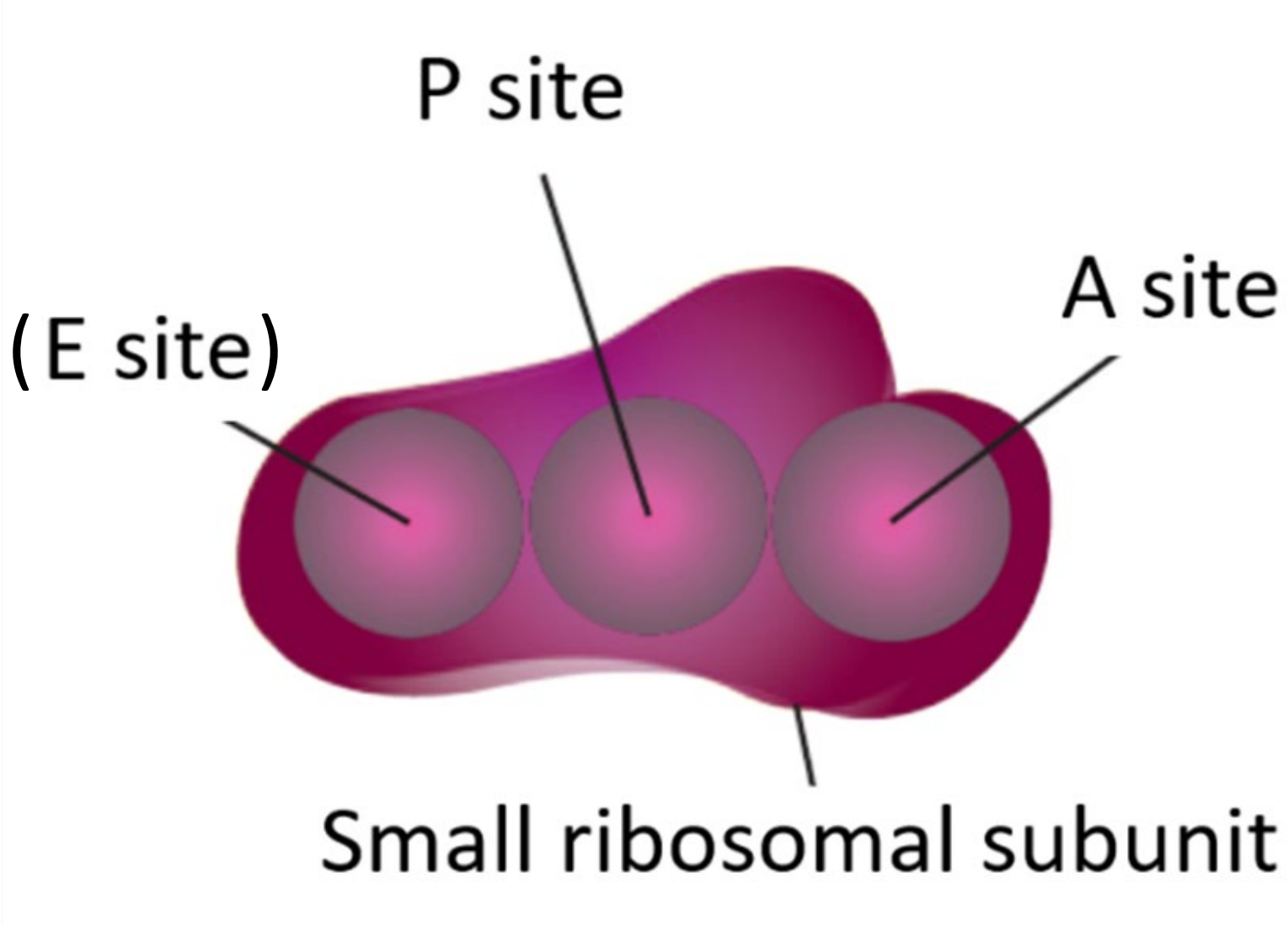
- The initiation codon is AUG (Met)
- Cells generally use distinct initiator and elongation tRNAs for methionine
- In mitochondria (as in prokaryotes), the methionine bound to the initiating tRNA is transformed into N-formylmethionine



Components required for initiation

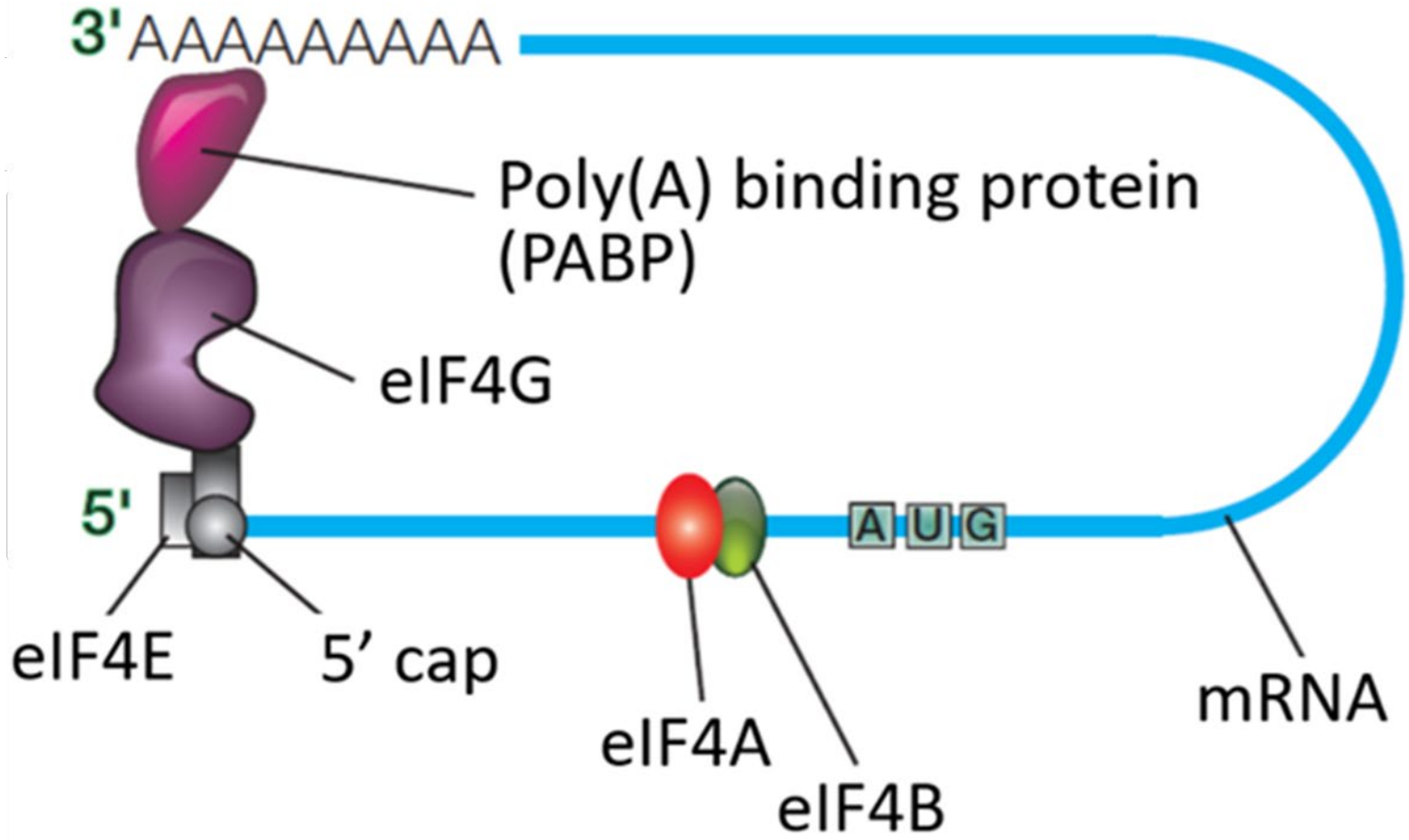
- Large and small ribosomal subunits
- Met-tRNA^{Met}
- mRNA
- At least 12 initiation factors (eIFs)
- GTP

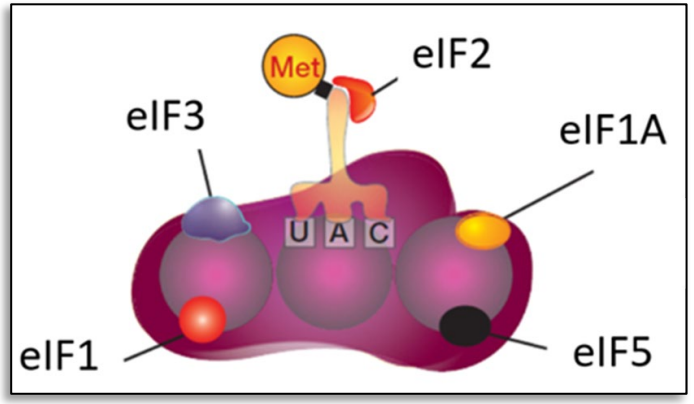
tRNA-binding sites in the ribosome



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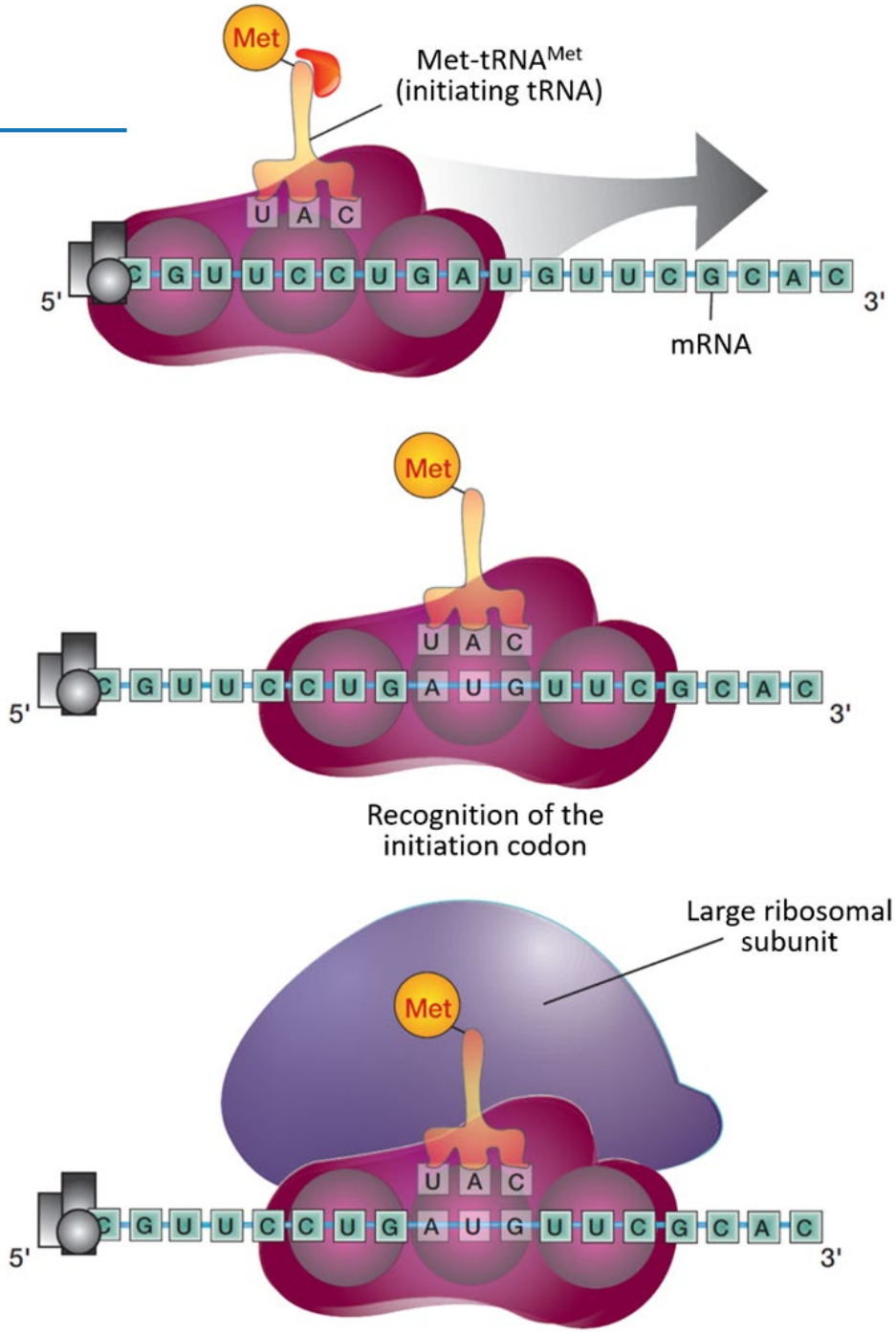
mRNA circularization and binding to the small ribosomal subunit





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Formation of the initiation complex



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Elongation

- Components:
  - Initiation complex
  - Aminoacyl-tRNAs
  - Elongation factors (eEFs)
  - GTP

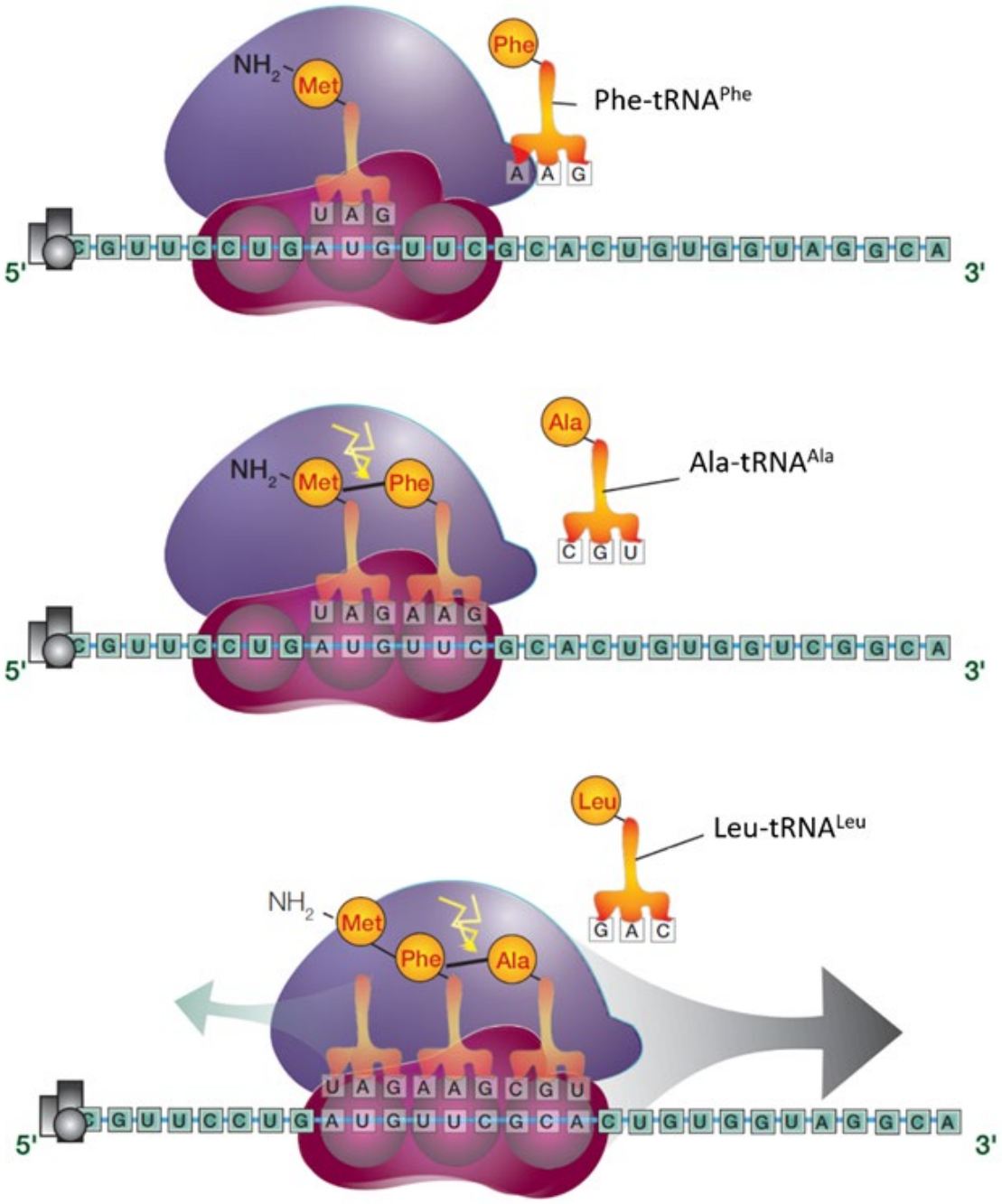
- Steps:
  - Binding of an incoming aminoacyl-tRNA
  - Formation of the peptide bond
  - Translocation

Elongation steps

Binding of the aminoacyl-tRNA

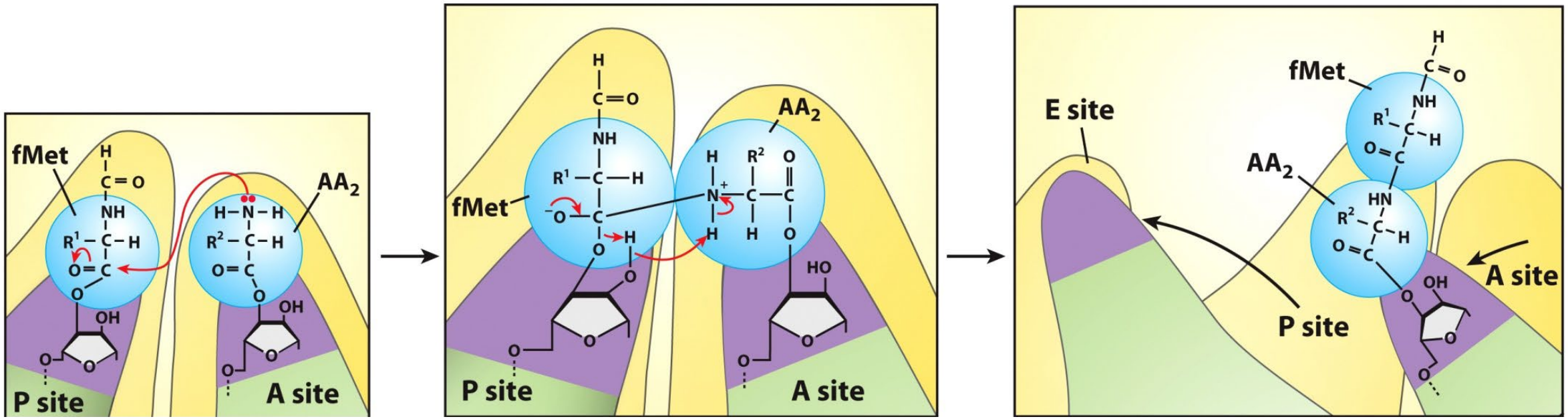
Formation of the peptide bond

Translocation



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Formation of the peptide bond

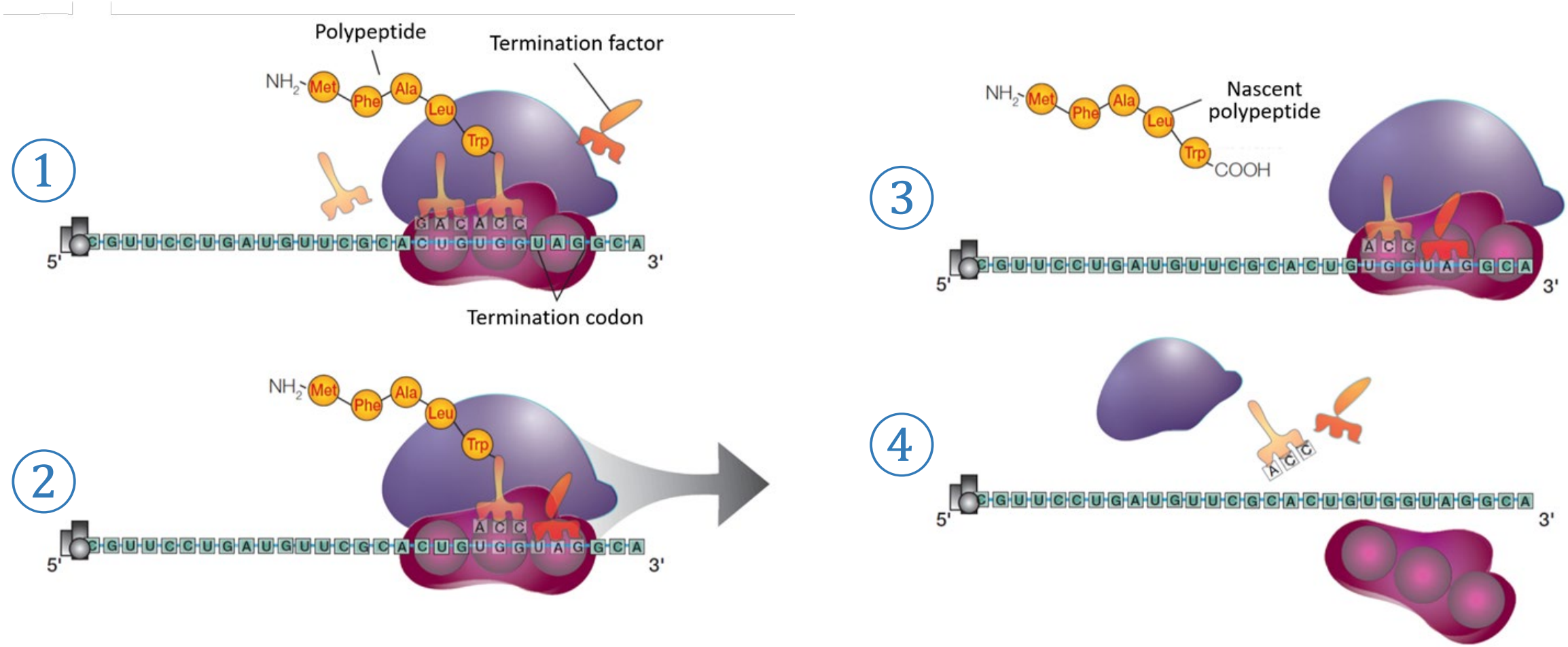


Peptidyl transferase

Peptidyl transferase is a ribozyme

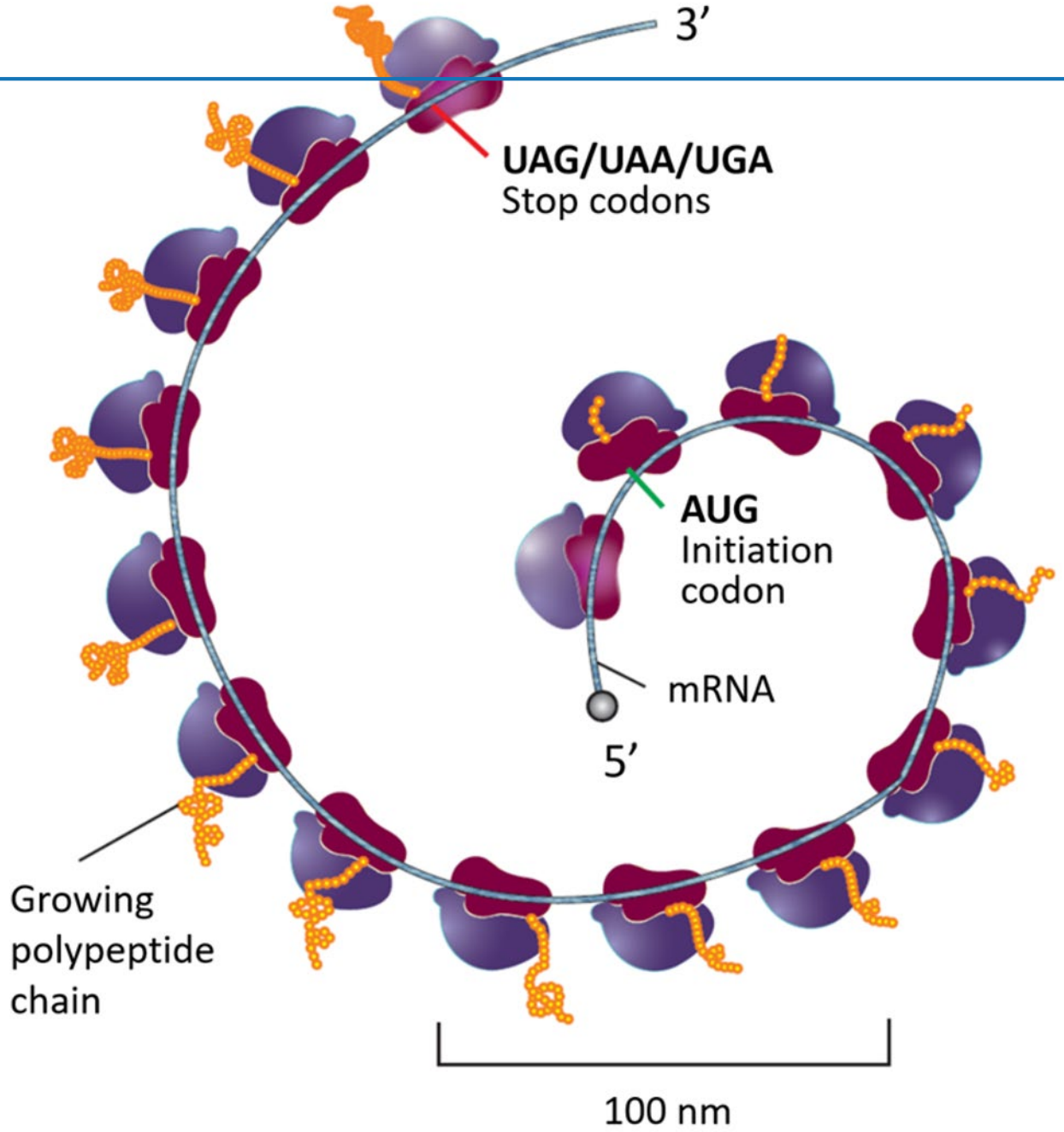
23S rRNA in bacteria and 28S rRNA in eukaryotes

Termination



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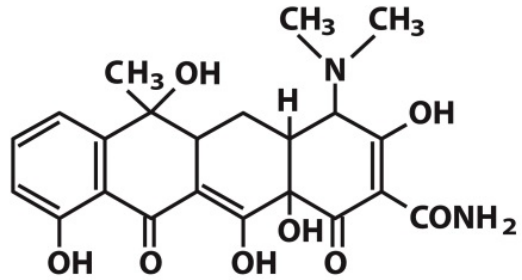
Polysomes



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Inhibition of translation

Antibiotics:



Tetracyclines: *block the A site on the ribosome*

Streptomycin: *causes misreading and inhibits initiation*

Chloramphenicol: inhibits the peptidyl transferase of 23S rRNA

Cycloheximide: *binds to 60S subunit and blocks translocation*

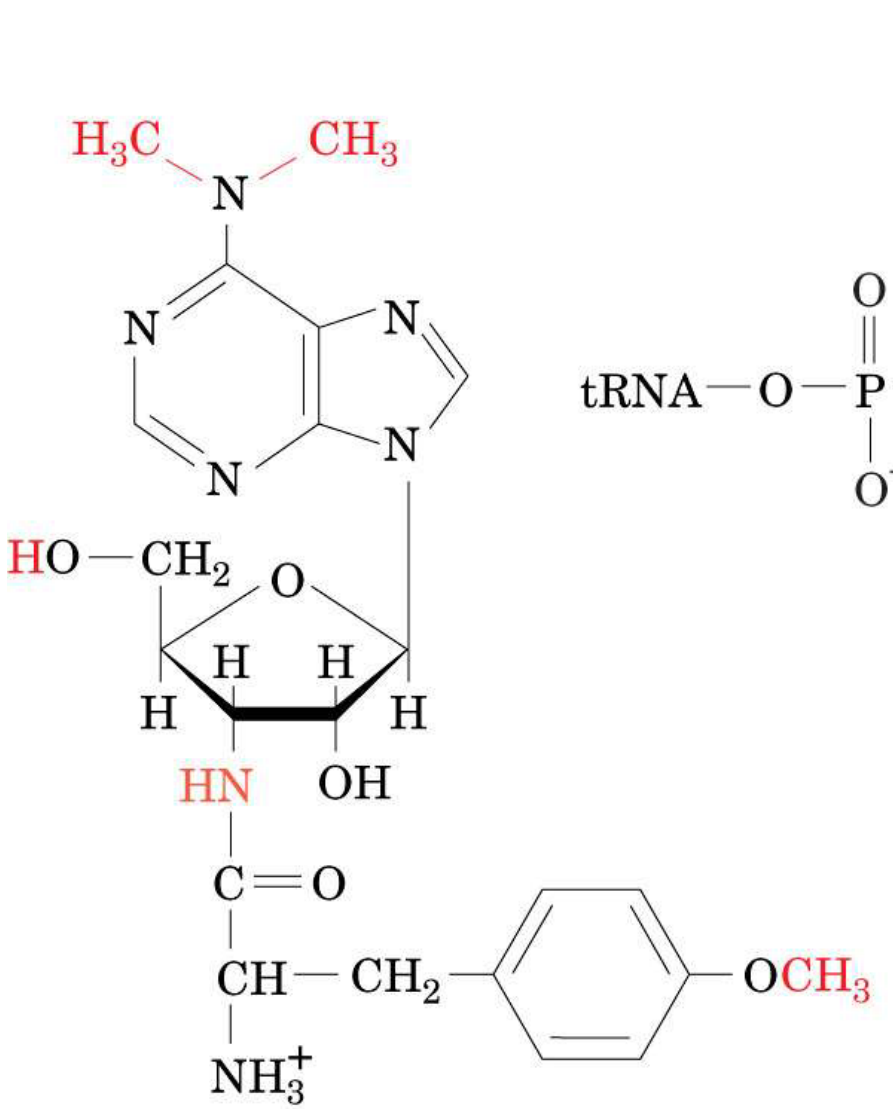
Puromycin: *terminates protein synthesis*

Toxins:

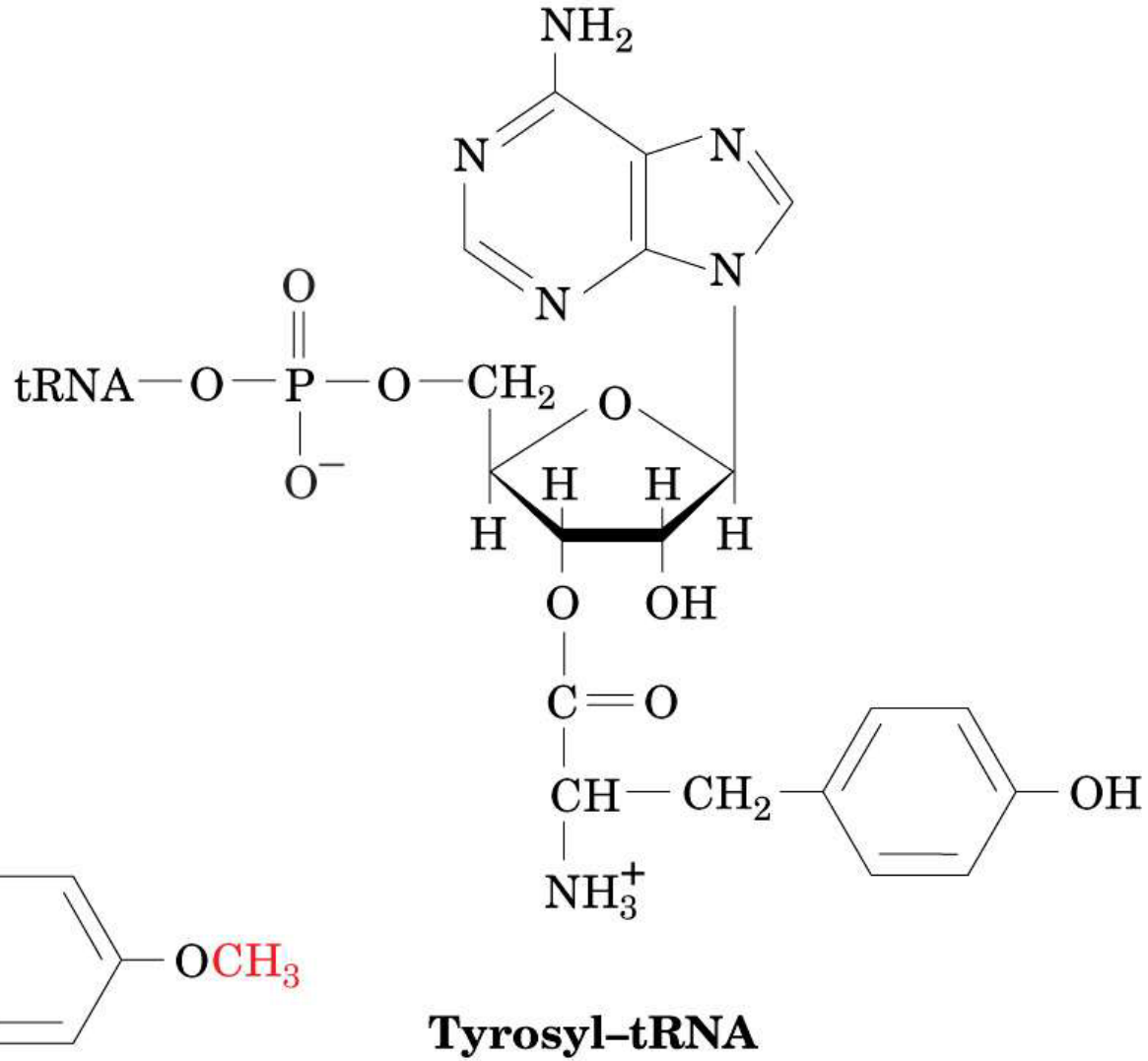
Diphtheria toxin: *inactivates eEF-2*

Ricin: *inactivates 28S rRNA*

Puromycin: a tRNA analog



Puromycin



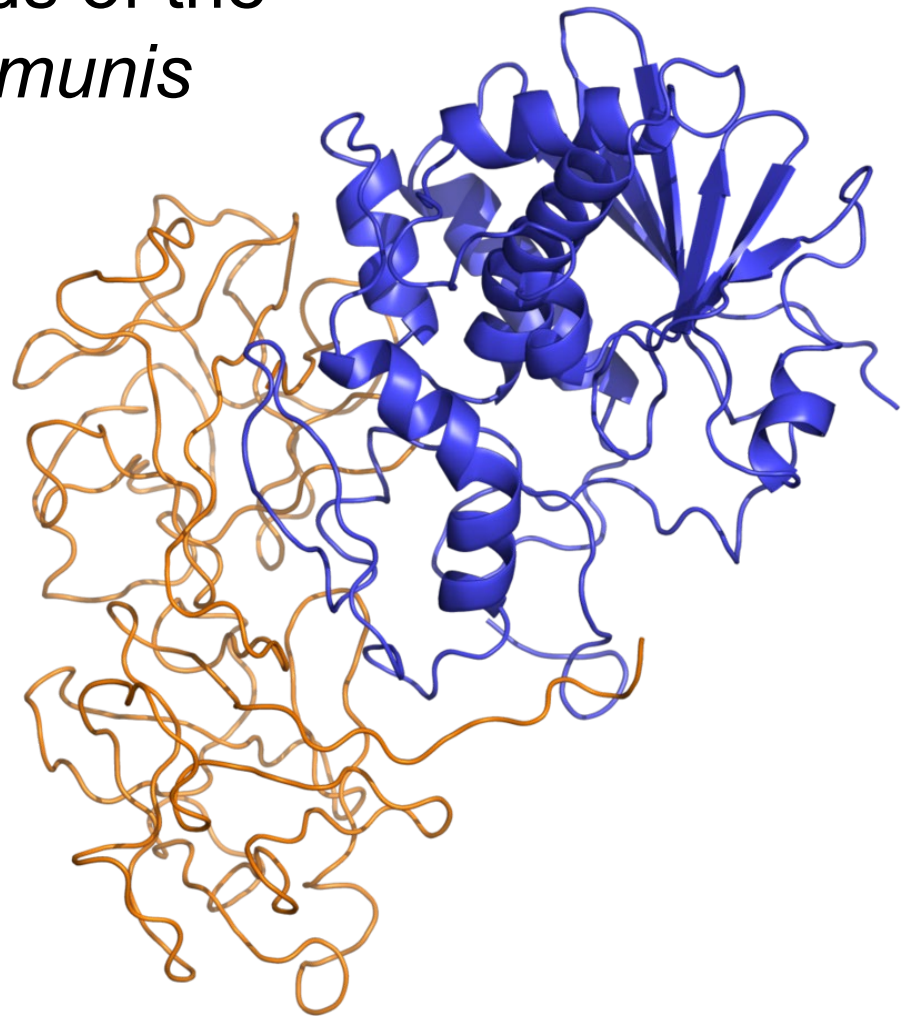
Tyrosyl–tRNA

Voet 3rd edition

Ricin: a very potent inhibitor of protein synthesis



Ricin is produced in the seeds of the castor oil plant *Ricinus communis*



The B chain is a lectin that binds to carbohydrates on the surface of eukaryotic cells

The A chain is an rRNA N-glycosylase that catalyzes the hydrolysis of an N-glycosidic bond of an adenine within the 28S rRNA